1.17 Proportion

Worked Solutions

1)

A appears to be a quadratic curve, so this would not represent $y \propto \frac{1}{x^3}$

B appears to be a cubic curve, so this would not represent $y \propto \frac{1}{x^3}$

C appears to be a reciprocal curve, and could represent $y \propto \frac{1}{x^3}$

D also appears to be a reciprocal curve, but would fit $y \propto \frac{1}{x^2}$ and not $y \propto \frac{1}{x^3}$

2 a)

$y \propto x^2 \Rightarrow y = kx^2$

2 b)

$y \propto \frac{1}{x} \Rightarrow y = \frac{k}{x}$

2 c)

$y \propto x^{\frac{1}{3}} \Rightarrow y = kx^{\frac{1}{3}}$

2 d)

$y \propto \frac{1}{(x+1)^3} \Rightarrow y = \frac{k}{(x+1)^3}$

3)

$$\begin{aligned} y \propto x^2 &\Rightarrow y = kx^2 \\ 100 = k \times 5^2 &\Rightarrow k = 4 \\ \therefore y &= 4x^2 \\ y &= 4 \times 10^2 = 400 \end{aligned}$$

4)

$$\begin{aligned} y \propto \frac{1}{x^{\frac{1}{4}}} &\Rightarrow y = \frac{k}{x^{\frac{1}{4}}} \\ 18 = \frac{k}{81^{\frac{1}{4}}} &\Rightarrow k = 54 \\ \therefore y &= \frac{54}{x^{\frac{1}{4}}} \\ 180 = \frac{54}{x^{\frac{1}{4}}} &\Rightarrow x^{\frac{1}{4}} = \frac{3}{10} \Rightarrow x = \frac{81}{10000} \end{aligned}$$

5)

$$V \propto T \Rightarrow V = kT$$
$$24 = k \times 300 \Rightarrow k = \frac{2}{25}$$
$$\therefore V = \frac{2}{25}T$$
$$V = \frac{2}{25} \times 400 = 32 \text{ litres}$$

6)

$$t \propto \frac{1}{n} \Rightarrow t = \frac{k}{n}$$
$$8 = \frac{k}{5} \Rightarrow k = 40$$
$$\therefore t = \frac{40}{n}$$
$$4 = \frac{40}{n} \Rightarrow n = 10 \text{ workers}$$

7)

$$R \propto C \Rightarrow R = kC$$
$$2 = k \times 0.5 \Rightarrow k = 4$$
$$\therefore R = 4C$$
$$6 = 4C \Rightarrow C = 1.5 \text{ mol/L}$$

8)

$$I \propto \frac{1}{d^2} \Rightarrow I = \frac{k}{d^2}$$
$$400 = \frac{k}{2^2} \Rightarrow k = 1600$$
$$\therefore I = \frac{1600}{d^2}$$
$$100 = \frac{1600}{d^2} \Rightarrow d^2 = 16 \Rightarrow d = 4 \text{ metres}$$

9)

$$y \propto x^{\frac{1}{2}} \Rightarrow y = ax^{\frac{1}{2}}$$

$$z \propto \frac{1}{y^{10}} \Rightarrow z = \frac{b}{y^{10}}$$

$$\therefore z = \frac{b}{(ax^{\frac{1}{2}})^{10}} \Rightarrow z = \frac{b}{a^{10}x^5} \Rightarrow z = \frac{k}{x^5}$$

$$2 = \frac{k}{4^5} \Rightarrow k = 2048$$

$$\therefore z = \frac{2048}{x^5}$$

$$\frac{1}{16} = \frac{2048}{x^5} \Rightarrow x^5 = 32768 \Rightarrow x = 8$$